

Linear Transformations

A **linear transformation**, $T: U \mapsto V$, is a function that carries elements of the vector space U (called the **domain**) to the vector space V (called the **codomain**), and which has two additional properties

1. $T(\mathbf{u}_1 + \mathbf{u}_2) = T(\mathbf{u}_1) + T(\mathbf{u}_2)$ for all $\mathbf{u}_1, \mathbf{u}_2 \in U$
2. $T(\alpha \mathbf{u}) = \alpha T(\mathbf{u})$ for all $\mathbf{u} \in U$ and all $\alpha \in \mathbb{C}$

(This definition contains Notation LT.)

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As every vector space property derives from vector addition and scalar multiplication, so too, every property of a linear transformation derives from these two defining properties.

$$\begin{array}{ccc}
 \mathbf{u}_1, \mathbf{u}_2 & \xrightarrow{T} & T(\mathbf{u}_1), T(\mathbf{u}_2) \\
 \downarrow + & & \downarrow + \\
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 \end{array}$$

Diagram DLTA. Definition of Linear Transformation, Additive

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 \downarrow \alpha & & \downarrow \alpha \\
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 \end{array}$$

Diagram DLTM. Definition of Linear Transformation, Multiplicative

Example

Define $T: \mathbb{C}^3 \mapsto \mathbb{C}^2$ by describing the output of the function for a generic input with the formula

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 + x_3 \\ -4x_2 \end{bmatrix}$$

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$$T(\mathbf{x} + \mathbf{y}) = T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \right)$$

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$$\begin{aligned} T(\mathbf{x} + \mathbf{y}) &= T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \right) \\ &= T \left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2(x_1 + y_1) + (x_3 + y_3) \\ -4(x_2 + y_2) \end{bmatrix} \\ &= \begin{bmatrix} (2x_1 + x_3) + (2y_1 + y_3) \\ -4x_2 + (-4)y_2 \end{bmatrix} \\ &= \begin{bmatrix} 2x_1 + x_3 \\ -4x_2 \end{bmatrix} + \begin{bmatrix} 2y_1 + y_3 \\ -4y_2 \end{bmatrix} \\ &= T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) + T \left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \right) \\ &= T(\mathbf{x}) + T(\mathbf{y}) \end{aligned}$$

and

$$\begin{aligned}T(\alpha \mathbf{x}) &= T\left(\alpha \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) \\&= T\left(\begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \alpha x_3 \end{bmatrix}\right) \\&= \begin{bmatrix} 2(\alpha x_1) + (\alpha x_3) \\ -4(\alpha x_2) \end{bmatrix} \\&= \begin{bmatrix} \alpha(2x_1 + x_3) \\ \alpha(-4x_2) \end{bmatrix} \\&= \alpha \begin{bmatrix} 2x_1 + x_3 \\ -4x_2 \end{bmatrix} \\&= \alpha T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) \\&= \alpha T(\mathbf{x})\end{aligned}$$

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So by [Definition LT](#) T is a linear transformation.

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Example NLT

Define $S: \mathbb{C}^3 \mapsto \mathbb{C}^3$ by

$$S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 4x_1 + 2x_2 \\ 0 \\ x_1 + 3x_3 - 2 \end{bmatrix}$$

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This function “looks” linear, but consider

$$3 S \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) = 3 \begin{bmatrix} 8 \\ 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 24 \end{bmatrix}$$

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So the second required property fails for the choice of $\alpha = 3$ and $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and by [Definition LT](#)

Linear transformation, polynomials to matrices

Define a linear transformation $T: P_3 \mapsto M_{22}$ by

$$T(a + bx + cx^2 + dx^3) = \begin{bmatrix} a + b & a - 2c \\ d & b - d \end{bmatrix}$$

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We verify the two defining conditions of a linear transformations.

$$\begin{aligned} T(\mathbf{x} + \mathbf{y}) &= T((a_1 + b_1x + c_1x^2 + d_1x^3) + (a_2 + b_2x + c_2x^2 + d_2x^3)) \\ &= T((a_1 + a_2) + (b_1 + b_2)x + (c_1 + c_2)x^2 + (d_1 + d_2)x^3) \\ &= \begin{bmatrix} (a_1 + a_2) + (b_1 + b_2) & (a_1 + a_2) - 2(c_1 + c_2) \\ d_1 + d_2 & (b_1 + b_2) - (d_1 + d_2) \end{bmatrix} \\ &= \begin{bmatrix} (a_1 + b_1) + (a_2 + b_2) & (a_1 - 2c_1) + (a_2 - 2c_2) \\ d_1 + d_2 & (b_1 - d_1) + (b_2 - d_2) \end{bmatrix} \\ &= \begin{bmatrix} a_1 + b_1 & a_1 - 2c_1 \\ d_1 & b_1 - d_1 \end{bmatrix} + \begin{bmatrix} a_2 + b_2 & a_2 - 2c_2 \\ d_2 & b_2 - d_2 \end{bmatrix} \\ &= T(a_1 + b_1x + c_1x^2 + d_1x^3) + T(a_2 + b_2x + c_2x^2 + d_2x^3) \\ &= T(\mathbf{x}) + T(\mathbf{y}) \end{aligned}$$

and

$$\begin{aligned}T(\alpha \mathbf{x}) &= T(\alpha(a + bx + cx^2 + dx^3)) \\&= T((\alpha a) + (\alpha b)x + (\alpha c)x^2 + (\alpha d)x^3) \\&= \begin{bmatrix} (\alpha a) + (\alpha b) & (\alpha a) - 2(\alpha c) \\ \alpha d & (\alpha b) - (\alpha d) \end{bmatrix} \\&= \begin{bmatrix} \alpha(a + b) & \alpha(a - 2c) \\ \alpha d & \alpha(b - d) \end{bmatrix} \\&= \alpha \begin{bmatrix} a + b & a - 2c \\ d & b - d \end{bmatrix}\end{aligned}$$

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Linear transformation, polynomials to polynomials

Define a function $S: P_4 \mapsto P_5$ by

$$S(p(x)) = (x - 2)p(x)$$

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Linear Transformations Take Zero to Zero

Suppose $T: U \mapsto V$ is a linear transformation. Then $T(\mathbf{0}) = \mathbf{0}$.



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Proof The two zero vectors in the conclusion of the theorem are different. The first is from U while the second is from V . We will subscript the zero vectors in this proof to highlight the distinction. Think about your objects.

$$\begin{aligned} T(\mathbf{0}_U) &= T(0\mathbf{0}_U) \\ &= 0T(\mathbf{0}_U) \\ &= \mathbf{0}_V \end{aligned}$$

Return to [Example NLT](#)

$$S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 4x_1 + 2x_2 \\ 0 \\ x_1 + 3x_3 - 2 \end{bmatrix}$$

and compute $S \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$

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compute $T(0 + 0x + 0x^2 + 0x^3) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ as an example of [Theorem](#)

T a linear transformation,

Matrices and Linear Transformations

Example

Linear transformation from a matrix

Let

$$A = \begin{bmatrix} 3 & -1 & 8 & 1 \\ 2 & 0 & 5 & -2 \\ 1 & 1 & 3 & -7 \end{bmatrix}$$

and define a function $P: \mathbb{C}^4 \mapsto \mathbb{C}^3$ by

$$P(\mathbf{x}) = A\mathbf{x}$$

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So we recognize the action of the function P as using the components of the vector (x_1, x_2, x_3, x_4) as scalars to form the output of P as a linear combination of the four columns of the matrix A , which are all members of $\mathbb{C}^3$, so the result is a vector in $\mathbb{C}^3$. We can rearrange this expression further, using our definitions of operations in $\mathbb{C}^3$

$$P(\mathbf{x}) = A\mathbf{x}$$

$$= x_1 \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 8 \\ 5 \\ 3 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -2 \\ -7 \end{bmatrix}$$

$$= \begin{bmatrix} 3x_1 \\ 2x_1 \\ x_1 \end{bmatrix} + \begin{bmatrix} -x_2 \\ 0 \\ x_2 \end{bmatrix} + \begin{bmatrix} 8x_3 \\ 5x_3 \\ 3x_3 \end{bmatrix} + \begin{bmatrix} x_4 \\ -2x_4 \\ -7x_4 \end{bmatrix}$$

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Theorem

Matrices Build Linear Transformations

Suppose that A is an $m \times n$ matrix. Define a function $T: \mathbb{C}^n \mapsto \mathbb{C}^m$ by $T(\mathbf{x}) = A\mathbf{x}$. Then T is a linear transformation. $\square$

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Proof

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Example

Matrix from a linear transformation

Define the function $R: \mathbb{C}^3 \mapsto \mathbb{C}^4$ by

$$R \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - 3x_2 + 4x_3 \\ x_1 + x_2 + x_3 \\ -x_1 + 5x_2 - 3x_3 \\ x_2 - 4x_3 \end{bmatrix}$$

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So if we define the matrix

$$B = \begin{bmatrix} 2 & -3 & 4 \\ 1 & 1 & 1 \\ -1 & 5 & -3 \\ 0 & 1 & -4 \end{bmatrix}$$

then $R(\mathbf{x}) = B\mathbf{x}$.

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Theorem

Matrix of a Linear Transformation, Column Vectors

Suppose that $T: \mathbb{C}^n \mapsto \mathbb{C}^m$ is a linear transformation. Then there is an $m \times n$ matrix A such that $T(\mathbf{x}) = A\mathbf{x}$. □

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Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n$ be the columns of the identity matrix of size n , I_n . Evaluate the linear transformation T with each of these standard unit vectors as an input, and record the result. In other words, define n vectors in $\mathbb{C}^m$, $\mathbf{A}_i$, $1 \leq i \leq n$ by

$$\mathbf{A}_i = T(\mathbf{e}_i)$$

Let $\mathbf{e}_j \in \mathbb{C}^m$, $1 \leq j \leq m$ denote the column vectors defined by

$$[\mathbf{e}_j]_i = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Then the set

$$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_m\} = \{\mathbf{e}_j \mid 1 \leq j \leq m\}$$

is the set of **standard unit vectors** in $\mathbb{C}^m$.

Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n$ be the columns of the identity matrix of size n , I_n . Evaluate the linear transformation T with each of these standard unit vectors as an input, and record the result. In other words, define n vectors in $\mathbb{C}^m$, $\mathbf{A}_i$, $1 \leq i \leq n$ by

$$\mathbf{A}_i = T(\mathbf{e}_i)$$

Let $\mathbf{e}_j \in \mathbb{C}^m$, $1 \leq j \leq m$ denote the column vectors defined by

$$[\mathbf{e}_j]_i = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Then the set

$$\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_m\} = \{\mathbf{e}_j \mid 1 \leq j \leq m\}$$

is the set of **standard unit vectors** in $\mathbb{C}^m$.

Then package up these vectors as the columns of a matrix

$$A = [\mathbf{A}_1 | \mathbf{A}_2 | \mathbf{A}_3 | \dots | \mathbf{A}_n]$$

$$\begin{aligned}
T(\mathbf{x}) &= T(I_n \mathbf{x}) \\
&= T([\mathbf{e}_1 | \mathbf{e}_2 | \mathbf{e}_3 | \dots | \mathbf{e}_n] \mathbf{x}) \\
&= T([\mathbf{x}]_1 \mathbf{e}_1 + [\mathbf{x}]_2 \mathbf{e}_2 + [\mathbf{x}]_3 \mathbf{e}_3 + \dots + [\mathbf{x}]_n \mathbf{e}_n) \\
&= T([\mathbf{x}]_1 \mathbf{e}_1) + T([\mathbf{x}]_2 \mathbf{e}_2) + T([\mathbf{x}]_3 \mathbf{e}_3) + \dots + T([\mathbf{x}]_n \mathbf{e}_n) \\
&= [\mathbf{x}]_1 T(\mathbf{e}_1) + [\mathbf{x}]_2 T(\mathbf{e}_2) + [\mathbf{x}]_3 T(\mathbf{e}_3) + \dots + [\mathbf{x}]_n T(\mathbf{e}_n) \\
&= [\mathbf{x}]_1 \mathbf{A}_1 + [\mathbf{x}]_2 \mathbf{A}_2 + [\mathbf{x}]_3 \mathbf{A}_3 + \dots + [\mathbf{x}]_n \mathbf{A}_n \\
&= \mathbf{A} \mathbf{x}
\end{aligned}$$

as desired.

$$\begin{aligned}
T(\mathbf{x}) &= T(I_n \mathbf{x}) \\
&= T([\mathbf{e}_1 | \mathbf{e}_2 | \mathbf{e}_3 | \dots | \mathbf{e}_n] \mathbf{x}) \\
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&= T([\mathbf{x}]_1 \mathbf{e}_1) + T([\mathbf{x}]_2 \mathbf{e}_2) + T([\mathbf{x}]_3 \mathbf{e}_3) + \dots + T([\mathbf{x}]_n \mathbf{e}_n) \\
&= [\mathbf{x}]_1 T(\mathbf{e}_1) + [\mathbf{x}]_2 T(\mathbf{e}_2) + [\mathbf{x}]_3 T(\mathbf{e}_3) + \dots + [\mathbf{x}]_n T(\mathbf{e}_n) \\
&= [\mathbf{x}]_1 \mathbf{A}_1 + [\mathbf{x}]_2 \mathbf{A}_2 + [\mathbf{x}]_3 \mathbf{A}_3 + \dots + [\mathbf{x}]_n \mathbf{A}_n \\
&= A\mathbf{x}
\end{aligned}$$

as desired.

Example

Matrix of a linear transformation

Suppose $S: \mathbb{C}^3 \mapsto \mathbb{C}^4$ is defined by

$$S\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 3x_1 - 2x_2 + 5x_3 \\ x_1 + x_2 + x_3 \\ 9x_1 - 2x_2 + 5x_3 \\ 4x_2 \end{bmatrix}$$

$$\begin{aligned}
T(\mathbf{x}) &= T(I_n \mathbf{x}) \\
&= T([\mathbf{e}_1 | \mathbf{e}_2 | \mathbf{e}_3 | \dots | \mathbf{e}_n] \mathbf{x}) \\
&= T([\mathbf{x}]_1 \mathbf{e}_1 + [\mathbf{x}]_2 \mathbf{e}_2 + [\mathbf{x}]_3 \mathbf{e}_3 + \dots + [\mathbf{x}]_n \mathbf{e}_n) \\
&= T([\mathbf{x}]_1 \mathbf{e}_1) + T([\mathbf{x}]_2 \mathbf{e}_2) + T([\mathbf{x}]_3 \mathbf{e}_3) + \dots + T([\mathbf{x}]_n \mathbf{e}_n) \\
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&= [\mathbf{x}]_1 \mathbf{A}_1 + [\mathbf{x}]_2 \mathbf{A}_2 + [\mathbf{x}]_3 \mathbf{A}_3 + \dots + [\mathbf{x}]_n \mathbf{A}_n \\
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Then

$$C_1 = S(e_1) = S \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 3 \\ 1 \\ 9 \\ 0 \end{bmatrix}$$

$$C_2 = S(e_2) = S \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -2 \\ 1 \\ -2 \\ 4 \end{bmatrix}$$

$$C_3 = S(e_3) = S \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ 1 \\ 5 \\ 0 \end{bmatrix}$$

so define

$$C = [C_1|C_2|C_3] = \begin{bmatrix} 3 & -2 & 5 \\ 1 & 1 & 1 \\ 9 & -2 & 5 \\ 0 & 4 & 0 \end{bmatrix}$$

and [Theorem](#) guarantees that $S(x) = Cx$.

As an illuminating exercise, let $\mathbf{z} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$ and compute $S(\mathbf{z})$ two different ways. First, return to the definition of S and evaluate $S(\mathbf{z})$ directly. Then do the matrix-vector product $C\mathbf{z}$. In both cases you should obtain the vector $S(\mathbf{z}) = \begin{bmatrix} 27 \\ 2 \\ 39 \\ -12 \end{bmatrix}$. ☒

Linear Transformations and Linear Combinations

Theorem

Linear Transformations and Linear Combinations

Suppose that $T: U \mapsto V$ is a linear transformation, $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_t$ are vectors from U and $a_1, a_2, a_3, \dots, a_t$ are scalars from $\mathbb{C}$. Then

$$T(a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3 + \cdots + a_t\mathbf{u}_t) = a_1T(\mathbf{u}_1) + a_2T(\mathbf{u}_2) + a_3T(\mathbf{u}_3) + \cdots + a_tT(\mathbf{u}_t)$$

□

Proof

$$\begin{aligned} T(a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3 + \cdots + a_t\mathbf{u}_t) \\ &= T(a_1\mathbf{u}_1) + T(a_2\mathbf{u}_2) + T(a_3\mathbf{u}_3) + \cdots + T(a_t\mathbf{u}_t) \\ &= a_1T(\mathbf{u}_1) + a_2T(\mathbf{u}_2) + a_3T(\mathbf{u}_3) + \cdots + a_tT(\mathbf{u}_t) \end{aligned}$$

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$$T(a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3 + \cdots + a_t\mathbf{u}_t) = a_1T(\mathbf{u}_1) + a_2T(\mathbf{u}_2) + a_3T(\mathbf{u}_3) + \cdots + a_tT(\mathbf{u}_t)$$

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Our next theorem says, informally, that it is enough to know how a linear transformation behaves for inputs from any basis of the domain, and *all* the other outputs are described by a linear combination of these few values.

Linear Transformation Defined on a Basis

Suppose $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ is a basis for the vector space U and $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n$ is a list of vectors from the vector space V (which are not necessarily distinct). Then there is a unique linear transformation, $T: U \mapsto V$, such that $T(\mathbf{u}_i) = \mathbf{v}_i$, $1 \leq i \leq n$. $\square$

there are unique scalars $a_1, a_2, a_3, \dots, a_n$ such that

$$\mathbf{w} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + a_3\mathbf{u}_3 + \cdots + a_n\mathbf{u}_n$$

$$T(\mathbf{w}) = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + a_3\mathbf{v}_3 + \cdots + a_n\mathbf{v}_n$$

Linear transformation defined on a basis

Consider the linear transformation $T: \mathbb{C}^3 \mapsto \mathbb{C}^2$ that is required to have the following three values,

$$T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad T \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \qquad T \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

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Because

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a basis for $\mathbb{C}^3$ there is a unique linear transformation T that behaves this way.

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is a basis for $\mathbb{C}^3$ there is a unique linear transformation T that behaves this way.

How do we compute other values of T ?

Consider the input

$$\mathbf{w} = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = (2) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (-3) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (1) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Then

$$T(\mathbf{w}) = (2) \begin{bmatrix} 2 \\ 1 \end{bmatrix} + (-3) \begin{bmatrix} -1 \\ 4 \end{bmatrix} + (1) \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 13 \\ -10 \end{bmatrix}$$

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Doing it again,

$$\mathbf{x} = \begin{bmatrix} 5 \\ 2 \\ -3 \end{bmatrix} = (5) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (2) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + (-3) \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

so

$$T(\mathbf{x}) = (5) \begin{bmatrix} 2 \\ 1 \end{bmatrix} + (2) \begin{bmatrix} -1 \\ 4 \end{bmatrix} + (-3) \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} -10 \\ 13 \end{bmatrix}$$

Example

Linear transformation defined on a basis

Consider the linear transformation $R: \mathbb{C}^3 \mapsto \mathbb{C}^2$ with the three values,

$$R\left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 5 \\ -1 \end{bmatrix} \qquad R\left(\begin{bmatrix} -1 \\ 5 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 4 \end{bmatrix} \qquad R\left(\begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

You can check that

$$D = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} \right\}$$

is a basis for $\mathbb{C}^3$ (make the vectors the columns of a square matrix and check that the matrix is nonsingular, By [Theorem](#) we know there is a unique linear transformation R with the three specified outputs. However, we have to work just a bit harder to take an input vector and express it as a linear combination of the vectors in D . For example, consider,

$$\mathbf{y} = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix}$$

Then we must first write $\mathbf{y}$ as a linear combination of the vectors in D and solve for the unknown scalars, to arrive at

$$\mathbf{y} = \begin{bmatrix} 8 \\ -3 \\ 5 \end{bmatrix} = (3) \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + (-2) \begin{bmatrix} -1 \\ 5 \\ 1 \end{bmatrix} + (1) \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$$

Then the proof of [Theorem](#) gives us

$$R(\mathbf{y}) = (3) \begin{bmatrix} 5 \\ -1 \end{bmatrix} + (-2) \begin{bmatrix} 0 \\ 4 \end{bmatrix} + (1) \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 17 \\ -8 \end{bmatrix}$$

Any other value of R could be computed in a similar manner.

☒

Linear transformation defined on a basis

The set $W = \{p(x) \in P_3 \mid p(1) = 0, p(3) = 0\} \subseteq P_3$ is a subspace of the vector space of polynomials P_3 . This subspace has $C = \{3 - 4x + x^2, 12 - 13x + x^3\}$ as a basis (check this!). Suppose we consider the linear transformation $S: P_3 \mapsto M_{22}$ with values

$$S(3 - 4x + x^2) = \begin{bmatrix} 1 & -3 \\ 2 & 0 \end{bmatrix}$$

$$S(12 - 13x + x^3) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Then the proof of [Theorem](#) gives us

$$R(\mathbf{y}) = (3) \begin{bmatrix} 5 \\ -1 \end{bmatrix} + (-2) \begin{bmatrix} 0 \\ 4 \end{bmatrix} + (1) \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 17 \\ -8 \end{bmatrix}$$

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By [Theorem](#) we know there is a unique linear transformation with these two values. To illustrate a sample computation of S , consider $q(x) = 9 - 6x - 5x^2 + 2x^3$. Verify that $q(x)$ is an element of W (does it have roots at $x = 1$ and $x = 3$?), then find the scalars needed to write it as a linear combination of the basis vectors in C . Because

$$q(x) = 9 - 6x - 5x^2 + 2x^3 = (-5)(3 - 4x + x^2) + (2)(12 - 13x + x^3)$$

The proof of [Theorem](#) gives us

$$S(q) = (-5) \begin{bmatrix} 1 & -3 \\ 2 & 0 \end{bmatrix} + (2) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 17 \\ -8 & 0 \end{bmatrix}$$

And all the other outputs of S could be computed in the same manner. Every output of S will have a

Pre-Image

Suppose that $T: U \mapsto V$ is a linear transformation. For each $\mathbf{v}$, define the **pre-image** of $\mathbf{v}$ to be the subset of U given by

$$T^{-1}(\mathbf{v}) = \{\mathbf{u} \in U \mid T(\mathbf{u}) = \mathbf{v}\}$$

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$$T^{-1}(\mathbf{v}) = \{\mathbf{u} \in U \mid T(\mathbf{u}) = \mathbf{v}\}$$

In other words, $T^{-1}(\mathbf{v})$ is the set of all those vectors in the domain U that get “sent” to the vector $\mathbf{v}$.

Example

Sample pre-images, .

is the linear transformation defined by

$$T: \mathbb{C}^3 \mapsto M_{22}, \quad T\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} a - b & 2a + 2b + c \\ 3a + b + c & -2a - 6b - 2c \end{bmatrix}$$

We could compute a pre-image for every element of the codomain M_{22} . However, even in a free textbook, we do not have the room to do that, so we will compute just two.

Choose

$$\mathbf{v} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \in M_{22}$$

for no particular reason. What is $T^{-1}(\mathbf{v})$? Suppose $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \in T^{-1}(\mathbf{v})$. The condition that $T(\mathbf{u}) = \mathbf{v}$ becomes

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} = \mathbf{v} = T(\mathbf{u}) = T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 - u_2 & 2u_1 + 2u_2 + u_3 \\ 3u_1 + u_2 + u_3 & -2u_1 - 6u_2 - 2u_3 \end{bmatrix}$$

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Using matrix equality ([Definition 1.1.1](#)), we arrive at a system of four equations in the three unknowns u_1, u_2, u_3 with an augmented matrix that we can row-reduce in the hunt for solutions,

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 2 & 2 & 1 & 1 \\ 3 & 1 & 1 & 3 \\ -2 & -6 & -2 & 2 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} \boxed{1} & 0 & \frac{1}{4} & \frac{5}{4} \\ 0 & \boxed{1} & \frac{1}{4} & -\frac{3}{4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We recognize this system as having infinitely many solutions described by the single free variable u_3 . Eventually obtaining the vector form of the solutions ([Theorem 1.1.1](#)) we can describe the preimage precisely as,

$$\begin{aligned}
T^{-1}(\mathbf{v}) &= \{ \mathbf{u} \in \mathbb{C}^3 \mid T(\mathbf{u}) = \mathbf{v} \} \\
&= \left\{ \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \mid u_1 = \frac{5}{4} - \frac{1}{4}u_3, u_2 = -\frac{3}{4} - \frac{1}{4}u_3 \right\} \\
&= \left\{ \begin{bmatrix} \frac{5}{4} - \frac{1}{4}u_3 \\ -\frac{3}{4} - \frac{1}{4}u_3 \\ u_3 \end{bmatrix} \mid u_3 \in \mathbb{C}^3 \right\} \\
&= \left\{ \begin{bmatrix} \frac{5}{4} \\ -\frac{3}{4} \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \\ 1 \end{bmatrix} \mid u_3 \in \mathbb{C}^3 \right\} \\
&= \begin{bmatrix} \frac{5}{4} \\ -\frac{3}{4} \\ 0 \end{bmatrix} + \left\langle \left\{ \begin{bmatrix} -\frac{1}{4} \\ -\frac{1}{4} \\ 1 \end{bmatrix} \right\} \right\rangle
\end{aligned}$$

This last line is merely a suggestive way of describing the set on the previous line.

OK, let's compute another preimage, but with a different outcome this time. Choose

$$\mathbf{v} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \in M_{22}$$

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By Theorem 1.1.2 we recognize this system as inconsistent. So no vector $\mathbf{u}$ is a member of $T^{-1}(\mathbf{v})$ and so

$$T^{-1}(\mathbf{v}) = \emptyset$$

□

Linear Transformation Addition

Suppose that $T: U \mapsto V$ and $S: U \mapsto V$ are two linear transformations with the same domain and codomain. Then their **sum** is the function $T + S: U \mapsto V$ whose outputs are defined by

$$(T + S)(\mathbf{u}) = T(\mathbf{u}) + S(\mathbf{u})$$

Sum of Linear Transformations is a Linear Transformation

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Suppose that $T: U \mapsto V$ and $S: U \mapsto V$ are two linear transformations with the same domain and codomain. Then $T + S: U \mapsto V$ is a linear transformation. $\square$

Proof

$$\begin{aligned}(T + S)(\mathbf{x} + \mathbf{y}) &= T(\mathbf{x} + \mathbf{y}) + S(\mathbf{x} + \mathbf{y}) \\ &= T(\mathbf{x}) + T(\mathbf{y}) + S(\mathbf{x}) + S(\mathbf{y}) \\ &= T(\mathbf{x}) + S(\mathbf{x}) + T(\mathbf{y}) + S(\mathbf{y}) \\ &= (T + S)(\mathbf{x}) + (T + S)(\mathbf{y})\end{aligned}$$

and

$$\begin{aligned}(T + S)(\alpha\mathbf{x}) &= T(\alpha\mathbf{x}) + S(\alpha\mathbf{x}) \\ &= \alpha T(\mathbf{x}) + \alpha S(\mathbf{x}) \\ &= \alpha(T(\mathbf{x}) + S(\mathbf{x})) \\ &= \alpha(T + S)(\mathbf{x})\end{aligned}$$

Example

Sum of two linear transformations

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^3$ and $S: \mathbb{C}^2 \mapsto \mathbb{C}^3$ are defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \end{bmatrix} \qquad S\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 4x_1 - x_2 \\ x_1 + 3x_2 \\ -7x_1 + 5x_2 \end{bmatrix}$$

Example

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Then

$$(T + S)\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) + S\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \end{bmatrix} + \begin{bmatrix} 4x_1 - x_2 \\ x_1 + 3x_2 \\ -7x_1 + 5x_2 \end{bmatrix} = \begin{bmatrix} 5x_1 + x_2 \\ 4x_1 - x_2 \\ -2x_1 + 7x_2 \end{bmatrix}$$

Example

Sum of two linear transformations

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^3$ and $S: \mathbb{C}^2 \mapsto \mathbb{C}^3$ are defined by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \end{bmatrix} \qquad S\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 4x_1 - x_2 \\ x_1 + 3x_2 \\ -7x_1 + 5x_2 \end{bmatrix}$$

Then

$$(T + S)\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) + S\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \end{bmatrix} + \begin{bmatrix} 4x_1 - x_2 \\ x_1 + 3x_2 \\ -7x_1 + 5x_2 \end{bmatrix} = \begin{bmatrix} 5x_1 + x_2 \\ 4x_1 - x_2 \\ -2x_1 + 7x_2 \end{bmatrix}$$

$T + S$ is also a linear transformation from $\mathbb{C}^2$ to $\mathbb{C}^3$.

Linear Transformation Scalar Multiplication

Suppose that $T: U \mapsto V$ is a linear transformation and $\alpha \in \mathbb{C}$. Then the **scalar multiple** is the function $\alpha T: U \mapsto V$ whose outputs are defined by

$$(\alpha T)(\mathbf{u}) = \alpha T(\mathbf{u})$$

Multiple of a Linear Transformation is a Linear Transformation

Suppose that $T: U \mapsto V$ is a linear transformation and $\alpha \in \mathbb{C}$. Then $(\alpha T): U \mapsto V$ is a linear transformation. $\square$

Multiple of a Linear Transformation is a Linear Transformation

Suppose that $T: U \mapsto V$ is a linear transformation and $\alpha \in \mathbb{C}$. Then $(\alpha T): U \mapsto V$ is a linear transformation. $\square$

$$\begin{aligned}(\alpha T)(\mathbf{x} + \mathbf{y}) &= \alpha(T(\mathbf{x} + \mathbf{y})) \\&= \alpha(T(\mathbf{x}) + T(\mathbf{y})) \\&= \alpha T(\mathbf{x}) + \alpha T(\mathbf{y}) \\&= (\alpha T)(\mathbf{x}) + (\alpha T)(\mathbf{y})\end{aligned}$$

and

$$\begin{aligned}(\alpha T)(\beta \mathbf{x}) &= \alpha T(\beta \mathbf{x}) \\&= \alpha(\beta T(\mathbf{x})) \\&= (\alpha\beta) T(\mathbf{x}) \\&= (\beta\alpha) T(\mathbf{x}) \\&= \beta(\alpha T(\mathbf{x})) \\&= \beta((\alpha T)(\mathbf{x}))\end{aligned}$$

Example

Suppose that $\tilde{T}: \mathbb{C}^4 \mapsto \mathbb{C}^3$ is defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 - x_3 + 2x_4 \\ x_1 + 5x_2 - 3x_3 + x_4 \\ -2x_1 + 3x_2 - 4x_3 + 2x_4 \end{bmatrix}$$

Example

Suppose that $T: \mathbb{C}^4 \mapsto \mathbb{C}^3$ is defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 - x_3 + 2x_4 \\ x_1 + 5x_2 - 3x_3 + x_4 \\ -2x_1 + 3x_2 - 4x_3 + 2x_4 \end{bmatrix}$$

For the sake of an example, choose $\alpha = 2$,

$$\alpha T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = 2T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = 2 \begin{bmatrix} x_1 + 2x_2 - x_3 + 2x_4 \\ x_1 + 5x_2 - 3x_3 + x_4 \\ -2x_1 + 3x_2 - 4x_3 + 2x_4 \end{bmatrix} = \begin{bmatrix} 2x_1 + 4x_2 - 2x_3 + 4x_4 \\ 2x_1 + 10x_2 - 6x_3 + 2x_4 \\ -4x_1 + 6x_2 - 8x_3 + 4x_4 \end{bmatrix}$$

Example

Suppose that $T: \mathbb{C}^4 \mapsto \mathbb{C}^3$ is defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 - x_3 + 2x_4 \\ x_1 + 5x_2 - 3x_3 + x_4 \\ -2x_1 + 3x_2 - 4x_3 + 2x_4 \end{bmatrix}$$

For the sake of an example, choose $\alpha = 2$,

$$\alpha T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = 2T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = 2 \begin{bmatrix} x_1 + 2x_2 - x_3 + 2x_4 \\ x_1 + 5x_2 - 3x_3 + x_4 \\ -2x_1 + 3x_2 - 4x_3 + 2x_4 \end{bmatrix} = \begin{bmatrix} 2x_1 + 4x_2 - 2x_3 + 4x_4 \\ 2x_1 + 10x_2 - 6x_3 + 2x_4 \\ -4x_1 + 6x_2 - 8x_3 + 4x_4 \end{bmatrix}$$

$2T$ is also a linear transformation from $\mathbb{C}^4$ to $\mathbb{C}^3$.

Linear Transformation Composition

Suppose that $T: U \mapsto V$ and $S: V \mapsto W$ are linear transformations. Then the **composition** of S and T is the function $(S \circ T): U \mapsto W$ whose outputs are defined by

$$(S \circ T)(\mathbf{u}) = S(T(\mathbf{u}))$$

$\triangle$

Given that T and S are linear transformations, it would be nice to know that $S \circ T$ is also a linear transformation.

Composition of Linear Transformations is a Linear Transformation

Suppose that $T: U \mapsto V$ and $S: V \mapsto W$ are linear transformations. Then $(S \circ T): U \mapsto W$ is a linear transformation. $\square$

Composition of Linear Transformations is a Linear Transformation

Suppose that $T: U \mapsto V$ and $S: V \mapsto W$ are linear transformations. Then $(S \circ T): U \mapsto W$ is a linear transformation. $\square$

$$\begin{aligned}(S \circ T)(\mathbf{x} + \mathbf{y}) &= S(T(\mathbf{x} + \mathbf{y})) \\ &= S(T(\mathbf{x}) + T(\mathbf{y})) \\ &= S(T(\mathbf{x})) + S(T(\mathbf{y})) \\ &= (S \circ T)(\mathbf{x}) + (S \circ T)(\mathbf{y})\end{aligned}$$

and

$$\begin{aligned}(S \circ T)(\alpha \mathbf{x}) &= S(T(\alpha \mathbf{x})) \\ &= S(\alpha T(\mathbf{x})) \\ &= \alpha S(T(\mathbf{x})) \\ &= \alpha(S \circ T)(\mathbf{x})\end{aligned}$$

Example

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^4$ and $S: \mathbb{C}^4 \mapsto \mathbb{C}^3$ are defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \qquad S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - x_2 + x_3 - x_4 \\ 5x_1 - 3x_2 + 8x_3 - 2x_4 \\ -4x_1 + 3x_2 - 4x_3 + 5x_4 \end{bmatrix}$$

Example

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^4$ and $S: \mathbb{C}^4 \mapsto \mathbb{C}^3$ are defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \qquad S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - x_2 + x_3 - x_4 \\ 5x_1 - 3x_2 + 8x_3 - 2x_4 \\ -4x_1 + 3x_2 - 4x_3 + 5x_4 \end{bmatrix}$$

$$\begin{aligned} (S \circ T) \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= S \left(T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) \right) \\ &= S \left(\begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2(x_1 + 2x_2) - (3x_1 - 4x_2) + (5x_1 + 2x_2) - (6x_1 - 3x_2) \\ 5(x_1 + 2x_2) - 3(3x_1 - 4x_2) + 8(5x_1 + 2x_2) - 2(6x_1 - 3x_2) \\ -4(x_1 + 2x_2) + 3(3x_1 - 4x_2) - 4(5x_1 + 2x_2) + 5(6x_1 - 3x_2) \end{bmatrix} \\ &= \begin{bmatrix} -2x_1 + 13x_2 \\ 24x_1 + 44x_2 \\ 15x_1 - 43x_2 \end{bmatrix} \end{aligned}$$

Example

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^4$ and $S: \mathbb{C}^4 \mapsto \mathbb{C}^3$ are defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \qquad S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - x_2 + x_3 - x_4 \\ 5x_1 - 3x_2 + 8x_3 - 2x_4 \\ -4x_1 + 3x_2 - 4x_3 + 5x_4 \end{bmatrix}$$

$$\begin{aligned} (S \circ T) \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= S \left(T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) \right) \\ &= S \left(\begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2(x_1 + 2x_2) - (3x_1 - 4x_2) + (5x_1 + 2x_2) - (6x_1 - 3x_2) \\ 5(x_1 + 2x_2) - 3(3x_1 - 4x_2) + 8(5x_1 + 2x_2) - 2(6x_1 - 3x_2) \\ -4(x_1 + 2x_2) + 3(3x_1 - 4x_2) - 4(5x_1 + 2x_2) + 5(6x_1 - 3x_2) \end{bmatrix} \\ &= \begin{bmatrix} -2x_1 + 13x_2 \\ 24x_1 + 44x_2 \\ 15x_1 - 43x_2 \end{bmatrix} \end{aligned}$$

$S \circ T$ is a linear transformation from $\mathbb{C}^2$ to $\mathbb{C}^3$.

Example

Suppose that $T: \mathbb{C}^2 \mapsto \mathbb{C}^4$ and $S: \mathbb{C}^4 \mapsto \mathbb{C}^3$ are defined by

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \qquad S \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - x_2 + x_3 - x_4 \\ 5x_1 - 3x_2 + 8x_3 - 2x_4 \\ -4x_1 + 3x_2 - 4x_3 + 5x_4 \end{bmatrix}$$

$$\begin{aligned} (S \circ T) \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) &= S \left(T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) \right) \\ &= S \left(\begin{bmatrix} x_1 + 2x_2 \\ 3x_1 - 4x_2 \\ 5x_1 + 2x_2 \\ 6x_1 - 3x_2 \end{bmatrix} \right) \\ &= \begin{bmatrix} 2(x_1 + 2x_2) - (3x_1 - 4x_2) + (5x_1 + 2x_2) - (6x_1 - 3x_2) \\ 5(x_1 + 2x_2) - 3(3x_1 - 4x_2) + 8(5x_1 + 2x_2) - 2(6x_1 - 3x_2) \\ -4(x_1 + 2x_2) + 3(3x_1 - 4x_2) - 4(5x_1 + 2x_2) + 5(6x_1 - 3x_2) \end{bmatrix} \\ &= \begin{bmatrix} -2x_1 + 13x_2 \\ 24x_1 + 44x_2 \\ 15x_1 - 43x_2 \end{bmatrix} \end{aligned}$$

$S \circ T$ is a linear transformation from $\mathbb{C}^2$ to $\mathbb{C}^3$.

homework

1. Is the function below a linear transformation? Why or why not?

$$T: \mathbb{C}^3 \mapsto \mathbb{C}^2, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 3x_1 - x_2 + x_3 \\ 8x_2 - 6 \end{bmatrix}$$

2. Determine the matrix representation of the linear transformation S below.

$$S: \mathbb{C}^2 \mapsto \mathbb{C}^3, \quad S \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} 3x_1 + 5x_2 \\ 8x_1 - 3x_2 \\ -4x_1 \end{bmatrix}$$

Injective Linear Transformation

Suppose $T: U \mapsto V$ is a linear transformation. Then T is **injective** if whenever $T(\mathbf{x}) = T(\mathbf{y})$, then $\mathbf{x} = \mathbf{y}$. $\triangle$

Injective Linear Transformation

Suppose $T: U \mapsto V$ is a linear transformation. Then T is **injective** if whenever $T(\mathbf{x}) = T(\mathbf{y})$, then $\mathbf{x} = \mathbf{y}$. $\triangle$

Given an arbitrary function, it is possible for two different inputs to yield the same output (think about the function $f(x) = x^2$ and the inputs $x = 3$ and $x = -3$). For an injective function, this never happens. If we have equal outputs ($T(\mathbf{x}) = T(\mathbf{y})$) then we must have achieved those equal outputs by employing equal inputs ($\mathbf{x} = \mathbf{y}$). Some authors prefer the term **one-to-one** where we use injective, and we will sometimes refer to an injective linear transformation as an **injection**.

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \right) = \begin{bmatrix} -2x_1 + 3x_2 + 3x_3 - 6x_4 + 3x_5 \\ -16x_1 + 9x_2 + 12x_3 - 28x_4 + 28x_5 \\ -19x_1 + 7x_2 + 14x_3 - 32x_4 + 37x_5 \\ -21x_1 + 9x_2 + 15x_3 - 35x_4 + 39x_5 \\ -9x_1 + 5x_2 + 7x_3 - 16x_4 + 16x_5 \end{bmatrix}$$

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{bmatrix} -2x_1 + 3x_2 + 3x_3 - 6x_4 + 3x_5 \\ -16x_1 + 9x_2 + 12x_3 - 28x_4 + 28x_5 \\ -19x_1 + 7x_2 + 14x_3 - 32x_4 + 37x_5 \\ -21x_1 + 9x_2 + 15x_3 - 35x_4 + 39x_5 \\ -9x_1 + 5x_2 + 7x_3 - 16x_4 + 16x_5 \end{bmatrix}$$

Notice that for

$$\mathbf{x} = \begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 4 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} 4 \\ 7 \\ 0 \\ 5 \\ 7 \end{bmatrix}$$

we have

$$T \left(\begin{bmatrix} 1 \\ 3 \\ -1 \\ 2 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} 4 \\ 55 \\ 72 \\ 77 \\ 31 \end{bmatrix} \qquad T \left(\begin{bmatrix} 4 \\ 7 \\ 0 \\ 5 \\ 7 \end{bmatrix} \right) = \begin{bmatrix} 4 \\ 55 \\ 72 \\ 77 \\ 31 \end{bmatrix}$$

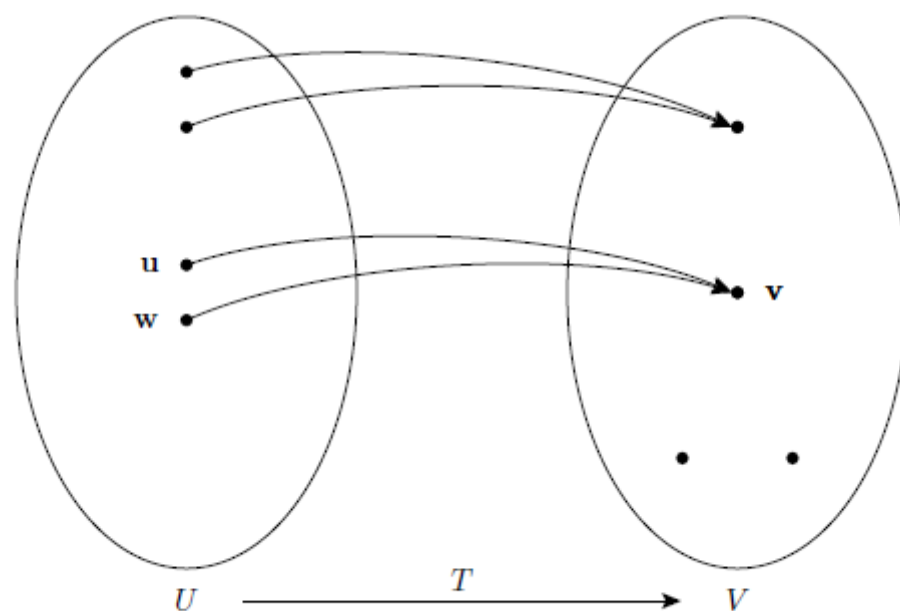


Diagram NILT. Non-Injective Linear Transformation

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \right) = \begin{bmatrix} -65x_1 + 128x_2 + 10x_3 - 262x_4 + 40x_5 \\ 36x_1 - 73x_2 - x_3 + 151x_4 - 16x_5 \\ -44x_1 + 88x_2 + 5x_3 - 180x_4 + 24x_5 \\ 34x_1 - 68x_2 - 3x_3 + 140x_4 - 18x_5 \\ 12x_1 - 24x_2 - x_3 + 49x_4 - 5x_5 \end{bmatrix}$$

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \right) = \begin{bmatrix} -65x_1 + 128x_2 + 10x_3 - 262x_4 + 40x_5 \\ 36x_1 - 73x_2 - x_3 + 151x_4 - 16x_5 \\ -44x_1 + 88x_2 + 5x_3 - 180x_4 + 24x_5 \\ 34x_1 - 68x_2 - 3x_3 + 140x_4 - 18x_5 \\ 12x_1 - 24x_2 - x_3 + 49x_4 - 5x_5 \end{bmatrix}$$

To establish that R is injective we must begin with the assumption that $T(\mathbf{x}) = T(\mathbf{y})$ and somehow arrive from this at the conclusion that $\mathbf{x} = \mathbf{y}$. Here we go,

$$T(\mathbf{x}) = T(\mathbf{y})$$

$$T \begin{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \end{pmatrix} = T \begin{pmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} \end{pmatrix}$$

$$\begin{bmatrix} -65x_1 + 128x_2 + 10x_3 - 262x_4 + 40x_5 \\ 36x_1 - 73x_2 - x_3 + 151x_4 - 16x_5 \\ -44x_1 + 88x_2 + 5x_3 - 180x_4 + 24x_5 \\ 34x_1 - 68x_2 - 3x_3 + 140x_4 - 18x_5 \\ 12x_1 - 24x_2 - x_3 + 49x_4 - 5x_5 \end{bmatrix} = \begin{bmatrix} -65y_1 + 128y_2 + 10y_3 - 262y_4 + 40y_5 \\ 36y_1 - 73y_2 - y_3 + 151y_4 - 16y_5 \\ -44y_1 + 88y_2 + 5y_3 - 180y_4 + 24y_5 \\ 34y_1 - 68y_2 - 3y_3 + 140y_4 - 18y_5 \\ 12y_1 - 24y_2 - y_3 + 49y_4 - 5y_5 \end{bmatrix}$$

$$\begin{bmatrix} -65x_1 + 128x_2 + 10x_3 - 262x_4 + 40x_5 \\ 36x_1 - 73x_2 - x_3 + 151x_4 - 16x_5 \\ -44x_1 + 88x_2 + 5x_3 - 180x_4 + 24x_5 \\ 34x_1 - 68x_2 - 3x_3 + 140x_4 - 18x_5 \\ 12x_1 - 24x_2 - x_3 + 49x_4 - 5x_5 \end{bmatrix} - \begin{bmatrix} -65y_1 + 128y_2 + 10y_3 - 262y_4 + 40y_5 \\ 36y_1 - 73y_2 - y_3 + 151y_4 - 16y_5 \\ -44y_1 + 88y_2 + 5y_3 - 180y_4 + 24y_5 \\ 34y_1 - 68y_2 - 3y_3 + 140y_4 - 18y_5 \\ 12y_1 - 24y_2 - y_3 + 49y_4 - 5y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -65(x_1 - y_1) + 128(x_2 - y_2) + 10(x_3 - y_3) - 262(x_4 - y_4) + 40(x_5 - y_5) \\ 36(x_1 - y_1) - 73(x_2 - y_2) - (x_3 - y_3) + 151(x_4 - y_4) - 16(x_5 - y_5) \\ -44(x_1 - y_1) + 88(x_2 - y_2) + 5(x_3 - y_3) - 180(x_4 - y_4) + 24(x_5 - y_5) \\ 34(x_1 - y_1) - 68(x_2 - y_2) - 3(x_3 - y_3) + 140(x_4 - y_4) - 18(x_5 - y_5) \\ 12(x_1 - y_1) - 24(x_2 - y_2) - (x_3 - y_3) + 49(x_4 - y_4) - 5(x_5 - y_5) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -65 & 128 & 10 & -262 & 40 \\ 36 & -73 & -1 & 151 & -16 \\ -44 & 88 & 5 & -180 & 24 \\ 34 & -68 & -3 & 140 & -18 \\ 12 & -24 & -1 & 49 & -5 \end{bmatrix} \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \\ x_3 - y_3 \\ x_4 - y_4 \\ x_5 - y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Now we recognize that we have a homogeneous system of 5 equations in 5 variables (the terms $x_i - y_i$ are the variables), so we row-reduce the coefficient matrix to

$$\begin{bmatrix} \boxed{1} & 0 & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

So the only solution is the trivial solution

$$x_1 - y_1 = 0 \qquad x_2 - y_2 = 0 \qquad x_3 - y_3 = 0 \qquad x_4 - y_4 = 0 \qquad x_5 - y_5 = 0$$

and we conclude that indeed $\mathbf{x} = \mathbf{y}$. By [Definition](#) T is injective.

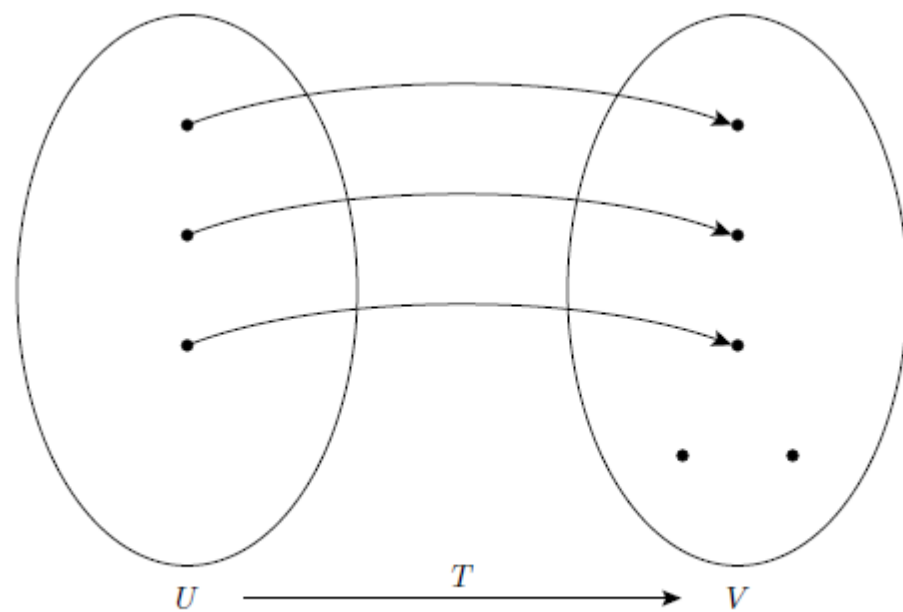


Diagram ILT. Injective Linear Transformation

Kernel of a Linear Transformation

For a linear transformation $T: U \mapsto V$, the kernel is a subset of the domain U . Informally, it is the set of all inputs that the transformation sends to the zero vector of the codomain.

Kernel of a Linear Transformation

Suppose $T: U \mapsto V$ is a linear transformation. Then the **kernel** of T is the set

$$\mathcal{K}(T) = \{\mathbf{u} \in U \mid T(\mathbf{u}) = \mathbf{0}\}$$

Notice that the kernel of T is just the preimage of $\mathbf{0}$, $T^{-1}(\mathbf{0})$

Example

the linear transformation

$$T: \mathbb{C}^3 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} -x_1 + x_2 - 3x_3 \\ -x_1 + 2x_2 - 4x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 3x_2 + x_3 \\ x_1 + 2x_3 \end{bmatrix}$$

Example

the linear transformation

$$T: \mathbb{C}^3 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} -x_1 + x_2 - 3x_3 \\ -x_1 + 2x_2 - 4x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 3x_2 + x_3 \\ x_1 + 2x_3 \end{bmatrix}$$

To determine the elements of $\mathbb{C}^3$ in $\mathcal{K}(T)$, find those vectors $\mathbf{u}$ such that $T(\mathbf{u}) = \mathbf{0}$, that is,

$$T(\mathbf{u}) = \mathbf{0}$$
$$\begin{bmatrix} -u_1 + u_2 - 3u_3 \\ -u_1 + 2u_2 - 4u_3 \\ u_1 + u_2 + u_3 \\ 2u_1 + 3u_2 + u_3 \\ u_1 + 2u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-u_1 + u_2 - 3u_3 = 0$$

$$-u_1 + 2u_2 - 4u_3 = 0$$

$$u_1 + u_2 + u_3 = 0$$

$$2u_1 + 3u_2 + u_3 = 0$$

$$u_1 + 2u_3 = 0$$

Row-reducing the coefficient matrix gives

$$\begin{bmatrix} \boxed{1} & 0 & 2 \\ 0 & \boxed{1} & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The kernel of T is the set of solutions to this homogeneous system of equations, which by [Theorem](#) can be expressed as

$$\mathcal{K}(T) = \left\langle \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\} \right\rangle$$

We know that the span of a set of vectors is always a subspace, so the kernel computed in [Example](#) is also a subspace. This is no accident, the kernel of a linear

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Kernel of a Linear Transformation is a Subspace

Suppose that $T: U \mapsto V$ is a linear transformation. Then the kernel of T , $\mathcal{K}(T)$, is a subspace of U . $\square$

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Suppose we assume that $\mathbf{x}, \mathbf{y} \in \mathcal{K}(T)$. Is $\mathbf{x} + \mathbf{y} \in \mathcal{K}(T)$?

$$\begin{aligned} T(\mathbf{x} + \mathbf{y}) &= T(\mathbf{x}) + T(\mathbf{y}) \\ &= \mathbf{0} + \mathbf{0} \\ &= \mathbf{0} \end{aligned}$$

This qualifies $\mathbf{x} + \mathbf{y}$ for membership in $\mathcal{K}(T)$. So we have additive closure.

Suppose we assume that $\alpha \in \mathbb{C}$ and $\mathbf{x} \in \mathcal{K}(T)$. Is $\alpha\mathbf{x} \in \mathcal{K}(T)$?

$$\begin{aligned} T(\alpha\mathbf{x}) &= \alpha T(\mathbf{x}) \\ &= \alpha \mathbf{0} \\ &= \mathbf{0} \end{aligned}$$

Example

linear transformation

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To determine the elements of $\mathbb{C}^3$ in $\mathcal{K}(T)$, find those vectors $\mathbf{u}$ such that $T(\mathbf{u}) = \mathbf{0}$, that is,

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$$\begin{aligned}
-u_1 + u_2 + u_3 &= 0 \\
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\end{aligned}$$

Row-reducing the coefficient matrix gives

$$\begin{bmatrix}
\boxed{1} & 0 & 0 \\
0 & \boxed{1} & 0 \\
0 & 0 & \boxed{1} \\
0 & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

The kernel of T is the set of solutions to this homogeneous system of equations, which is simply the trivial solution $\mathbf{u} = \mathbf{0}$, so

$$\mathcal{K}(T) = \{\mathbf{0}\} = \langle \{ \} \rangle$$



Kernel and Pre-Image

Suppose $T: U \mapsto V$ is a linear transformation and $\mathbf{v} \in V$. If the preimage $T^{-1}(\mathbf{v})$ is non-empty, and $\mathbf{u} \in T^{-1}(\mathbf{v})$ then

$$T^{-1}(\mathbf{v}) = \{\mathbf{u} + \mathbf{z} \mid \mathbf{z} \in \mathcal{K}(T)\} = \mathbf{u} + \mathcal{K}(T)$$

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Proof Let $M = \{\mathbf{u} + \mathbf{z} \mid \mathbf{z} \in \mathcal{K}(T)\}$. First, we show that $M \subseteq T^{-1}(\mathbf{v})$. Suppose that $\mathbf{w} \in M$, so $\mathbf{w}$ has the form $\mathbf{w} = \mathbf{u} + \mathbf{z}$, where $\mathbf{z} \in \mathcal{K}(T)$. Then

$$\begin{aligned} T(\mathbf{w}) &= T(\mathbf{u} + \mathbf{z}) \\ &= T(\mathbf{u}) + T(\mathbf{z}) \\ &= \mathbf{v} + \mathbf{0} \\ &= \mathbf{v} \end{aligned}$$

which qualifies $\mathbf{w}$ for membership in the preimage of $\mathbf{v}$, $\mathbf{w} \in T^{-1}(\mathbf{v})$.

For the opposite inclusion, suppose $\mathbf{x} \in T^{-1}(\mathbf{v})$. Then,

$$\begin{aligned} T(\mathbf{x} - \mathbf{u}) &= T(\mathbf{x}) - T(\mathbf{u}) \\ &= \mathbf{v} - \mathbf{v} \\ &= \mathbf{0} \end{aligned}$$

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Suppose $T: U \mapsto V$ is a linear transformation and $\mathbf{v} \in V$. If the preimage $T^{-1}(\mathbf{v})$ is non-empty, and $\mathbf{u} \in T^{-1}(\mathbf{v})$ then

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Proof Let $M = \{\mathbf{u} + \mathbf{z} \mid \mathbf{z} \in \mathcal{K}(T)\}$. First, we show that $M \subseteq T^{-1}(\mathbf{v})$. Suppose that $\mathbf{w} \in M$, so $\mathbf{w}$ has the form $\mathbf{w} = \mathbf{u} + \mathbf{z}$, where $\mathbf{z} \in \mathcal{K}(T)$. Then

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which qualifies $\mathbf{w}$ for membership in the preimage of $\mathbf{v}$, $\mathbf{w} \in T^{-1}(\mathbf{v})$.

For the opposite inclusion, suppose $\mathbf{x} \in T^{-1}(\mathbf{v})$. Then,

$$\begin{aligned} T(\mathbf{x} - \mathbf{u}) &= T(\mathbf{x}) - T(\mathbf{u}) \\ &= \mathbf{v} - \mathbf{v} \\ &= \mathbf{0} \end{aligned}$$

This qualifies $\mathbf{x} - \mathbf{u}$ for membership in the kernel of T , $\mathcal{K}(T)$. So there is a vector $\mathbf{z} \in \mathcal{K}(T)$ such that $\mathbf{x} - \mathbf{u} = \mathbf{z}$. Rearranging this equation gives $\mathbf{x} = \mathbf{u} + \mathbf{z}$ and so $\mathbf{x} \in M$. So $T^{-1}(\mathbf{v}) \subseteq M$ and we see that $M = T^{-1}(\mathbf{v})$, as desired. ■

Kernel of an Injective Linear Transformation

Suppose that $T: U \mapsto V$ is a linear transformation. Then T is injective if and only if the kernel of T is trivial, $\mathcal{K}(T) = \{\mathbf{0}\}$. $\square$

Since the kernel is a subspace $\{\mathbf{0}\} \subseteq \mathcal{K}(T)$. To establish the opposite inclusion, suppose $\mathbf{x} \in \mathcal{K}(T)$.

$$\begin{aligned} T(\mathbf{x}) &= \mathbf{0} \\ &= T(\mathbf{0}) \end{aligned}$$

| to conclude that $\mathbf{x} = \mathbf{0}$. Therefore $\mathcal{K}(T) \subseteq \{\mathbf{0}\}$ and by Definition $\mathcal{K}(T) = \{\mathbf{0}\}$.

($\Leftarrow$) To establish that T is injective, appeal to Definition | and begin with the assumption that $T(\mathbf{x}) = T(\mathbf{y})$. Then

$$\begin{aligned} T(\mathbf{x} - \mathbf{y}) &= T(\mathbf{x}) - T(\mathbf{y}) \\ &= \mathbf{0} \end{aligned}$$

So $\mathbf{x} - \mathbf{y} \in \mathcal{K}(T)$ by Definition KLT and with the hypothesis that the kernel is trivial we conclude that $\mathbf{x} - \mathbf{y} = \mathbf{0}$. Then

$$\mathbf{y} = \mathbf{y} + \mathbf{0} = \mathbf{y} + (\mathbf{x} - \mathbf{y}) = \mathbf{x}$$

thus establishing that T is injective by Definition

Example

the kernel was determined to be

$$\left\langle \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\} \right\rangle$$

a subspace of $\mathbb{C}^3$ with dimension 1. Since the kernel is not trivial, [Theorem](#) tells us that T is not injective.

Injective Linear Transformations and Linear Independence

Suppose that $T: U \mapsto V$ is an injective linear transformation and $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_t\}$ is a linearly independent subset of U . Then $R = \{T(\mathbf{u}_1), T(\mathbf{u}_2), T(\mathbf{u}_3), \dots, T(\mathbf{u}_t)\}$ is a linearly independent subset of V . $\square$

Injective Linear Transformations and Bases

Suppose that $T: U \mapsto V$ is a linear transformation and $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\}$ is a basis of U . Then T is injective if and only if $C = \{T(\mathbf{u}_1), T(\mathbf{u}_2), T(\mathbf{u}_3), \dots, T(\mathbf{u}_m)\}$ is a linearly independent subset of V . $\square$

Injective Linear Transformations and Dimension

Suppose that $T: U \mapsto V$ is an injective linear transformation. Then $\dim(U) \leq \dim(V)$. □

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The linear transformation

$$T: M_{23} \mapsto \mathbb{C}^4, \quad T\left(\begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}\right) = \begin{bmatrix} a + 2b + 12c - 3d + e + 6f \\ 2a - b - c + d - 11f \\ a + b + 7c + 2d + e - 3f \\ a + 2b + 12c + 5e - 5f \end{bmatrix}$$

Since $\dim(M_{23}) = 6 > 4 = \dim(\mathbb{C}^4)$, T cannot be injective

Surjective Linear Transformation

Suppose $T: U \mapsto V$ is a linear transformation. Then T is **surjective** if for every $\mathbf{v} \in V$ there exists a $\mathbf{u} \in U$ so that $T(\mathbf{u}) = \mathbf{v}$. $\triangle$

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \right) = \begin{bmatrix} -2x_1 + 3x_2 + 3x_3 - 6x_4 + 3x_5 \\ -16x_1 + 9x_2 + 12x_3 - 28x_4 + 28x_5 \\ -19x_1 + 7x_2 + 14x_3 - 32x_4 + 37x_5 \\ -21x_1 + 9x_2 + 15x_3 - 35x_4 + 39x_5 \\ -9x_1 + 5x_2 + 7x_3 - 16x_4 + 16x_5 \end{bmatrix}$$

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We will demonstrate that

$$\mathbf{v} = \begin{bmatrix} -1 \\ 2 \\ 3 \\ -1 \\ 4 \end{bmatrix}$$

is an unobtainable element of the codomain. Suppose to the contrary that $\mathbf{u}$ is an element of the

domain such that $T(\mathbf{u}) = \mathbf{v}$. Then

$$\begin{aligned} \begin{bmatrix} -1 \\ 2 \\ 3 \\ -1 \\ 4 \end{bmatrix} &= \mathbf{v} = T(\mathbf{u}) = T \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} \right) \\ &= \begin{bmatrix} -2u_1 + 3u_2 + 3u_3 - 6u_4 + 3u_5 \\ -16u_1 + 9u_2 + 12u_3 - 28u_4 + 28u_5 \\ -19u_1 + 7u_2 + 14u_3 - 32u_4 + 37u_5 \\ -21u_1 + 9u_2 + 15u_3 - 35u_4 + 39u_5 \\ -9u_1 + 5u_2 + 7u_3 - 16u_4 + 16u_5 \end{bmatrix} \\ &= \begin{bmatrix} -2 & 3 & 3 & -6 & 3 \\ -16 & 9 & 12 & -28 & 28 \\ -19 & 7 & 14 & -32 & 37 \\ -21 & 9 & 15 & -35 & 39 \\ -9 & 5 & 7 & -16 & 16 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} \end{aligned}$$

Now we recognize the appropriate input vector $\mathbf{u}$ as a solution to a linear system of equations. Form the augmented matrix of the system, and row-reduce to

$$\left[\begin{array}{cccccc} \boxed{1} & 0 & 0 & 0 & -1 & 0 \\ 0 & \boxed{1} & 0 & 0 & -\frac{4}{3} & 0 \\ 0 & 0 & \boxed{1} & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} \end{array} \right]$$

the system is inconsistent.

Example

$$T: \mathbb{C}^5 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \right) = \begin{bmatrix} -65x_1 + 128x_2 + 10x_3 - 262x_4 + 40x_5 \\ 36x_1 - 73x_2 - x_3 + 151x_4 - 16x_5 \\ -44x_1 + 88x_2 + 5x_3 - 180x_4 + 24x_5 \\ 34x_1 - 68x_2 - 3x_3 + 140x_4 - 18x_5 \\ 12x_1 - 24x_2 - x_3 + 49x_4 - 5x_5 \end{bmatrix}$$

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To establish that R is surjective we must begin with a totally arbitrary element of the codomain, $\mathbf{v}$ and somehow find an input vector $\mathbf{u}$ such that $T(\mathbf{u}) = \mathbf{v}$. We desire,

$$T(\mathbf{u}) = \mathbf{v}$$

Example

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We recognize this equation as a system of equations in the variables u_i , but our vector of constants contains symbols. In general, we would have to row-reduce the augmented matrix by hand, due to the symbolic final column. However, in this particular example, the 5×5 coefficient matrix is nonsingular we now know that T is surjective.

Example

$$T: P_3 \mapsto M_{22}, \quad T(a + bx + cx^2 + dx^3) = \begin{bmatrix} a + b & a - 2c \\ d & b - d \end{bmatrix}$$

To establish that the linear transformation is surjective, begin by choosing an arbitrary output. In this example, we need to choose an arbitrary 2×2 matrix, say

$$\mathbf{v} = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$$

and we would like to find an input polynomial

$$\mathbf{u} = a + bx + cx^2 + dx^3$$

so that $T(\mathbf{u}) = \mathbf{v}$. So we have,

$$\begin{aligned}\begin{bmatrix} x & y \\ z & w \end{bmatrix} &= \mathbf{v} \\ &= T(\mathbf{u}) \\ &= T(a + bx + cx^2 + dx^3) \\ &= \begin{bmatrix} a + b & a - 2c \\ d & b - d \end{bmatrix}\end{aligned}$$

Matrix equality leads us to the system of four equations in the four unknowns, x, y, z, w ,

$$\begin{aligned}a + b &= x \\ a - 2c &= y \\ d &= z \\ b - d &= w\end{aligned}$$

which can be rewritten as a matrix equation,

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

The coefficient matrix is nonsingular, hence it has an inverse,

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

so we have

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 0 & 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \\ = \begin{bmatrix} x - z - w \\ z + w \\ \frac{1}{2}(x - y - z - w) \\ z \end{bmatrix}$$

So the input polynomial $\mathbf{u} = (x - z - w) + (z + w)x + \frac{1}{2}(x - y - z - w)x^2 + zx^3$ will yield the output matrix $\mathbf{v}$, no matter what form $\mathbf{v}$ takes. This means by [Definition SLT \[516\]](#) that T is surjective. All the same, let's do a concrete demonstration and evaluate T with $\mathbf{u}$,

$$T(\mathbf{u}) = T\left((x - z - w) + (z + w)x + \frac{1}{2}(x - y - z - w)x^2 + zx^3\right)$$

$$\begin{aligned}
&= \begin{bmatrix} (x-z-w) + (z+w) & (x-z-w) - 2(\frac{1}{2}(x-y-z-w)) \\ z & (z+w) - z \end{bmatrix} \\
&= \begin{bmatrix} x & y \\ z & w \end{bmatrix} \\
&= \mathbf{v}
\end{aligned}$$

Burda kaldık

Range of a Linear Transformation

Suppose $T: U \mapsto V$ is a linear transformation. Then the **range** of T is the set

$$\mathcal{R}(T) = \{T(\mathbf{u}) \mid \mathbf{u} \in U\}$$

Example

$$T: \mathbb{C}^3 \mapsto \mathbb{C}^5, \quad T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} -x_1 + x_2 - 3x_3 \\ -x_1 + 2x_2 - 4x_3 \\ x_1 + x_2 + x_3 \\ 2x_1 + 3x_2 + x_3 \\ x_1 + 2x_3 \end{bmatrix}$$

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To determine the elements of $\mathbb{C}^5$ in $\mathcal{R}(T)$, find those vectors $\mathbf{v}$ such that $T(\mathbf{u}) = \mathbf{v}$ for some $\mathbf{u} \in \mathbb{C}^3$,

$$\begin{aligned} \mathbf{v} &= T(\mathbf{u}) \\ &= \begin{bmatrix} -u_1 + u_2 - 3u_3 \\ -u_1 + 2u_2 - 4u_3 \\ u_1 + u_2 + u_3 \\ 2u_1 + 3u_2 + u_3 \\ u_1 + 2u_3 \end{bmatrix} \\ &= \begin{bmatrix} -u_1 \\ -u_1 \\ u_1 \\ 2u_1 \\ u_1 \end{bmatrix} + \begin{bmatrix} u_2 \\ 2u_2 \\ u_2 \\ 3u_2 \\ 0 \end{bmatrix} + \begin{bmatrix} -3u_3 \\ -4u_3 \\ u_3 \\ u_3 \\ 2u_3 \end{bmatrix} \end{aligned}$$

$$= u_1 \begin{bmatrix} -1 \\ -1 \\ 1 \\ 2 \\ 1 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} -3 \\ -4 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

This says that every output of T ($\mathbf{v}$) can be written as a linear combination of the three vectors

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -3 \\ -4 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

using the scalars u_1, u_2, u_3 . Furthermore, since $\mathbf{u}$ can be any element of $\mathbb{C}^3$, every such linear combination is an output. This means that

$$\mathcal{R}(T) = \left\langle \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ -4 \\ 1 \\ 1 \\ 2 \end{bmatrix} \right\} \right\rangle$$

The three vectors in this spanning set for $\mathcal{R}(T)$ form a linearly dependent set

Let $L: R^3 \rightarrow R^3$ be defined by

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

- (a) Is L onto?
- (b) Find a basis for range L .
- (c) Find $\ker L$.
- (d) Is L one-to-one?

Solution

(a) Given any $\mathbf{w} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in R^3 , where a , b , and c are any real numbers, can we

find $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ so that $L(\mathbf{v}) = \mathbf{w}$? We seek a solution to the linear system

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix},$$

and we find the reduced row echelon form of the augmented matrix to be (verify)

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & a \\ 0 & 1 & 1 & b-a \\ 0 & 0 & 0 & c-b-a \end{array} \right].$$

Thus a solution exists only for $c - b - a = 0$, so L is not onto.

(b) To find a basis for range L , we note that

$$\begin{aligned} L\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}\right) &= \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} v_1 + v_3 \\ v_1 + v_2 + 2v_3 \\ 2v_1 + v_2 + 3v_3 \end{bmatrix} \\ &= v_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + v_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + v_3 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}. \end{aligned}$$

This means that

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right\}$$

spans range L . That is, range L is the subspace of R^3 spanned by the columns of the matrix defining L .

The first two vectors in this set are linearly independent, since they are not constant multiples of each other. The third vector is the sum of the first two. Therefore, the first two vectors form a basis for range L , and $\dim \text{range } L = 2$.

(c) To find $\ker L$, we wish to find all $\mathbf{v}$ in R^3 so that $L(\mathbf{v}) = \mathbf{0}_{R^3}$. Solving the resulting homogeneous system, we find (verify) that $v_1 = -v_3$ and $v_2 = -v_3$. Thus $\ker L$ consists of all vectors of the form

$$\begin{bmatrix} -a \\ -a \\ a \end{bmatrix} = a \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix},$$

where a is any real number. Moreover, $\dim \ker L = 1$.

(d) Since $\ker L \neq \{\mathbf{0}_{R^3}\}$, it follows from Theorem 6.4(b) that L is not one-to-one.

Let $L: R_4 \rightarrow R_3$ be defined by

$$L \left(\begin{bmatrix} u_1 & u_2 & u_3 & u_4 \end{bmatrix} \right) = \begin{bmatrix} u_1 + u_2 & u_3 + u_4 & u_1 + u_3 \end{bmatrix}.$$

Find a basis for range L .

We have

$$L([u_1 \ u_2 \ u_3 \ u_4]) = u_1[1 \ 0 \ 1] + u_2[1 \ 0 \ 0] \\ + u_3[0 \ 1 \ 1] + u_4[0 \ 1 \ 0].$$

Thus

$$S = \{[1 \ 0 \ 1], [1 \ 0 \ 0], [0 \ 1 \ 1], [0 \ 1 \ 0]\}$$

spans range L . To find a subset of S that is a basis for range L , we proceed as in Theorem 4.9 by first writing

$$u_1[1 \ 0 \ 1] + u_2[1 \ 0 \ 0] + u_3[0 \ 1 \ 1] + u_4[0 \ 1 \ 0] = [0 \ 0 \ 0].$$

The reduced row echelon form of the augmented matrix of this homogeneous system is (verify)

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right].$$

Since the leading 1's appear in columns 1, 2, and 3, we conclude that the first three vectors in S form a basis for range L . Thus

$$\{[1 \ 0 \ 1], [1 \ 0 \ 0], [0 \ 1 \ 1]\}$$

is a basis for range L .

To determine if a linear transformation is one-to-one or onto, we must solve a linear system. This is one further demonstration of the frequency with which linear systems must be solved to answer many questions in linear algebra. Finally, from Example 7, where $\dim \ker L = 1$, $\dim \text{range } L = 2$, and $\dim \text{domain } L = 3$, we saw that

$$\dim \ker L + \dim \text{range } L = \dim \text{domain } L.$$

This very important result is always true, and we now prove it in the following theorem:

Theorem

If $L: V \rightarrow W$ is a linear transformation of an n -dimensional vector space V into a vector space W , then

$$\dim \ker L + \dim \operatorname{range} L = \dim V.$$

Let $L: R^3 \rightarrow R^3$ be the linear transformation defined by

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} u_1 + u_3 \\ u_1 + u_2 \\ u_2 - u_3 \end{bmatrix}.$$

A vector $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ is in $\ker L$ if

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We must then find a basis for the solution space of the homogeneous system

$$\begin{aligned} u_1 + u_3 &= 0 \\ u_1 + u_2 &= 0 \\ u_2 - u_3 &= 0. \end{aligned}$$

We find (verify) that a basis for $\ker L$ is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$, so $\dim \ker L = 1$, and $\ker L$ is a line through the origin.

Next, every vector in range L is of the form

$$\begin{bmatrix} u_1 + u_3 \\ u_1 + u_2 \\ u_2 - u_3 \end{bmatrix},$$

which can be written as

$$u_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + u_3 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

Then a basis for range L is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(explain), so $\dim \text{range } L = 2$ and range L is a plane passing through the origin. These results are illustrated in Figure 6.1. Moreover,

$$\dim R^3 = 3 = \dim \ker L + \dim \text{range } L = 1 + 2,$$

EXAMPLE

Let $L: P_2 \rightarrow P_2$ be the linear transformation defined by

$$L(at^2 + bt + c) = (a + 2b)t + (b + c).$$

- (a) Find a basis for $\ker L$.
- (b) Find a basis for $\text{range } L$.

(a) The vector $at^2 + bt + c$ is in $\ker L$ if

$$L(at^2 + bt + c) = \mathbf{0},$$

that is, if

$$(a + 2b)t + (b + c) = 0.$$

Then

$$\begin{aligned}a + 2b &= 0 \\ b + c &= 0.\end{aligned}$$

Transforming the augmented matrix of this linear system to reduced row echelon form, we find that a basis for the solution space, and then a basis for $\ker L$, is (verify)

$$\{2t^2 - t + 1\}.$$

(b) Every vector in $\text{range } L$ has the form

$$(a + 2b)t + (b + c),$$

so the vectors t and 1 span $\text{range } L$. Since these vectors are also linearly independent, they form a basis for $\text{range } L$.

(c) From (a), $\dim \ker L = 1$, and from (b), $\dim \text{range } L = 2$, so

$$\dim \ker L + \dim \text{range } L = \dim P_2 = 3.$$



If $L: V \rightarrow W$ is a linear transformation of a vector space V into a vector space W and $\dim V = \dim W$, then the following statements are true:

- (a) If L is one-to-one, then it is onto.
- (b) If L is onto, then it is one-to-one.

A linear transformation $L : V \rightarrow W$ is invertible if and only if L is one-to-one and onto. Moreover, L^{-1} is a linear transformation and $(L^{-1})^{-1} = L$.

Consider the linear operator $L: R^3 \rightarrow R^3$ defined by

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

Consider the linear operator $L: R^3 \rightarrow R^3$ defined by

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

Since $\ker L = \{\mathbf{0}\}$ (verify), L is one-to-one, and it is also onto, so it is invertible. To obtain L^{-1} , we proceed as follows. Since $L^{-1}(\mathbf{w}) = \mathbf{v}$, we must solve $L(\mathbf{v}) = \mathbf{w}$ for $\mathbf{v}$. We have

$$L(\mathbf{v}) = L \left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \right) = \begin{bmatrix} v_1 + v_2 + v_3 \\ 2v_1 + 2v_2 + v_3 \\ v_2 + v_3 \end{bmatrix} = \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}.$$

We are then solving the linear system

$$\begin{aligned} v_1 + v_2 + v_3 &= w_1 \\ 2v_1 + 2v_2 + v_3 &= w_2 \\ v_2 + v_3 &= w_3 \end{aligned}$$

for v_1 , v_2 , and v_3 . We find that (verify)

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \mathbf{v} = L^{-1}(\mathbf{w}) = L^{-1} \left(\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \right) = \begin{bmatrix} w_1 - w_3 \\ -2w_1 + w_2 + w_3 \\ 2w_1 - w_2 \end{bmatrix}. \quad \blacksquare$$

Let $L: V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W ($n \neq 0, m \neq 0$) and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ be ordered bases for V and W , respectively. Then the $m \times n$ matrix A whose j th column is the coordinate vector $[L(\mathbf{v}_j)]_T$ of $L(\mathbf{v}_j)$

with respect to T has the following property:

$$[L(\mathbf{x})]_T = A [\mathbf{x}]_S \quad \text{for every } \mathbf{x} \text{ in } V.$$

Moreover, A is the only matrix with this property.

EXAMPLE

Let $L: P_2 \rightarrow P_1$ be defined by $L(p(t)) = p'(t)$, and consider the ordered bases $S = \{t^2, t, 1\}$ and $T = \{t, 1\}$ for P_2 and P_1 , respectively.

- (a) Find the matrix A associated with L .
- (b) If $p(t) = 5t^2 - 3t + 2$, compute $L(p(t))$ directly and then by using A .

Solution

(a) We have

$$L(t^2) = 2t = 2t + 0(1), \quad \text{so } [L(t^2)]_T = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

$$L(t) = 1 = 0(t) + 1(1), \quad \text{so } [L(t)]_T = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$L(1) = 0 = 0(t) + 0(1), \quad \text{so } [L(1)]_T = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

In this case, the coordinates of $L(t^2)$, $L(t)$, and $L(1)$ with respect to the T -basis are obtained by observation, since the T -basis is quite simple. Thus

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

(b) Since $p(t) = 5t^2 - 3t + 2$, then $L(p(t)) = 10t - 3$. However, we can find $L(p(t))$ by using the matrix A as follows: Since

$$[p(t)]_S = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix},$$

then

$$[L(p(t))]_T = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 10 \\ -3 \end{bmatrix},$$

which means that $L(p(t)) = 10t - 3$.

Let $L: P_2 \rightarrow P_1$ be defined as in Example 1 and consider the ordered bases $S = \{1, t, t^2\}$ and $T = \{t, 1\}$ for P_2 and P_1 , respectively. We then find that the matrix A associated with L is $\begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}$ (verify). Notice that if we change the order of the vectors in S or T , the matrix may change. ■

Let $L: P_2 \rightarrow P_1$ be defined as in Example 1, and consider the ordered bases $S = \{t^2, t, 1\}$ and $T = \{t + 1, t - 1\}$ for P_2 and P_1 , respectively.

- (a) Find the matrix A associated with L .
- (b) If $p(t) = 5t^2 - 3t + 2$, compute $L(p(t))$.

Let $L: P_2 \rightarrow P_1$ be defined as in Example 1, and consider the ordered bases $S = \{t^2, t, 1\}$ and $T = \{t + 1, t - 1\}$ for P_2 and P_1 , respectively.

- (a) Find the matrix A associated with L .
- (b) If $p(t) = 5t^2 - 3t + 2$, compute $L(p(t))$.

(a) We have

$$L(t^2) = 2t.$$

To find the coordinates of $L(t^2)$ with respect to the T -basis, we form

$$L(t^2) = 2t = a_1(t + 1) + a_2(t - 1),$$

which leads to the linear system

$$a_1 + a_2 = 2$$

$$a_1 - a_2 = 0,$$

whose solution is $a_1 = 1$, $a_2 = 1$ (verify). Hence

$$[L(t^2)]_T = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Similarly,

$$L(t) = 1 = \frac{1}{2}(t + 1) - \frac{1}{2}(t - 1), \quad \text{so } [L(t)]_T = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}.$$

$$L(1) = 0 = 0(t + 1) + 0(t - 1), \quad \text{so } [L(1)]_T = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Hence $A = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \end{bmatrix}$.

(b) We have

$$[L(p(t))]_T = \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 1 & -\frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{7}{2} \\ \frac{13}{2} \end{bmatrix},$$

so $L(p(t)) = \frac{7}{2}(t+1) + \frac{13}{2}(t-1) = 10t - 3$, which agrees with the result found in Example 1. ■

EXAMPLE

Let $L: R^3 \rightarrow R^2$ be defined by

$$L\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Let

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$\bar{\mathbf{e}}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \bar{\mathbf{e}}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then $S = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $T = \{\bar{\mathbf{e}}_1, \bar{\mathbf{e}}_2\}$ are the natural bases for R^3 and R^2 , respectively.

Now

$$L(\mathbf{e}_1) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\bar{\mathbf{e}}_1 + 1\bar{\mathbf{e}}_2, \quad \text{so } [L(\mathbf{e}_1)]_T = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

$$L(\mathbf{e}_2) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1\bar{\mathbf{e}}_1 + 2\bar{\mathbf{e}}_2, \quad \text{so } [L(\mathbf{e}_2)]_T = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$L(\mathbf{e}_3) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} = 1\bar{\mathbf{e}}_1 + 3\bar{\mathbf{e}}_2, \quad \text{so } [L(\mathbf{e}_3)]_T = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

In this case, the coordinate vectors of $L(\mathbf{e}_1)$, $L(\mathbf{e}_2)$, and $L(\mathbf{e}_3)$ with respect to the T -basis are readily computed, because T is the natural basis for R^2 . Then the representation of L with respect to S and T is

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}.$$

The reason that A is the same matrix as the one involved in the definition of L is that the natural bases are being used for R^3 and R^2 . ■

EXAMPLE

Let $L: R^3 \rightarrow R^2$ be defined as in Example 4, and consider the ordered bases

$$S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \text{and} \quad T = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

for R^3 and R^2 , respectively. Then

$$L \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Similarly,

$$L \left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \text{and} \quad L \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

To determine the coordinates of the images of the S -basis, we must solve the three linear systems

$$a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \mathbf{b},$$

where $\mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

by transforming the partitioned matrix

$$\left[\begin{array}{cc|cc|c} 1 & 1 & 2 & 2 & 1 \\ 2 & 3 & 3 & 5 & 3 \end{array} \right]$$

by transforming the partitioned matrix

$$\left[\begin{array}{cc|cc|c} 1 & 1 & 2 & 2 & 1 \\ 2 & 3 & 3 & 5 & 3 \end{array} \right]$$

to reduced row echelon form, yielding (verify)

$$\left[\begin{array}{cc|cc|c} 1 & 0 & 3 & 1 & 0 \\ 0 & 1 & -1 & 1 & 1 \end{array} \right].$$

The last three columns of this matrix are the desired coordinate vectors of the image of the S -basis with respect to the T -basis. That is, the last three columns form the matrix A representing L with respect to S and T . Thus

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

This matrix, of course, differs from the one that defined L . Thus, although a matrix A may be involved in the definition of a linear transformation L , we cannot conclude that it is necessarily the representation of L that we seek. ■

We can summarize the conditions under which a linear transformation L of an n -dimensional vector space V into itself (or, more generally, into an n -dimensional vector space W) is invertible by the following equivalent statements:

1. L is invertible.
2. L is one-to-one.
3. L is onto.
4. The matrix A representing L with respect to ordered bases S and T for V and W is nonsingular.

EXAMPLE

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}$, and let $S = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $T = \{\bar{\mathbf{e}}_1, \bar{\mathbf{e}}_2\}$ be the natural bases for R^3 and R^2 , respectively.

- (a) Find the unique linear transformation $L: R^3 \rightarrow R^2$ whose representation with respect to S and T is A .
- (b) Let

$$S' = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \quad \text{and} \quad T' = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}$$

be ordered bases for R^3 and R^2 , respectively. Determine the linear transformation $L: R^3 \rightarrow R^2$ whose representation with respect to S' and T' is A .

- (c) Compute $L \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right)$, using L as determined in part (b).

Solution

(a) Let

$$L(\mathbf{e}_1) = 1\bar{\mathbf{e}}_1 + 2\bar{\mathbf{e}}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

$$L(\mathbf{e}_2) = 2\bar{\mathbf{e}}_1 - 1\bar{\mathbf{e}}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix},$$

$$L(\mathbf{e}_3) = -\bar{\mathbf{e}}_1 + 3\bar{\mathbf{e}}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

Now if $\mathbf{x} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ is in R^3 , we define $L(\mathbf{x})$ by

$$\begin{aligned} L(\mathbf{x}) &= L(a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3) \\ &= a_1L(\mathbf{e}_1) + a_2L(\mathbf{e}_2) + a_3L(\mathbf{e}_3), \end{aligned}$$

$$\begin{aligned} L(\mathbf{x}) &= a_1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + a_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} a_1 + 2a_2 - a_3 \\ 2a_1 - a_2 + 3a_3 \end{bmatrix}. \end{aligned}$$

Note that

$$L(\mathbf{x}) = \begin{bmatrix} 1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix},$$

so we could have defined L by $L(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x}$ in R^3 . We can do this when the bases S and T are the natural bases.

(b) Let

$$L\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) = 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix},$$

$$L\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \end{bmatrix},$$

$$L\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right) = -1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -6 \end{bmatrix}.$$

Then if $\mathbf{x} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$, we express $\mathbf{x}$ in terms of the basis S' as

$$\mathbf{x} = b_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + b_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b_3 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}; \quad \text{hence } [\mathbf{x}]_{S'} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Define $L(\mathbf{x})$ by

$$L(\mathbf{x}) = b_1 L\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) + b_2 L\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) + b_3 L\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right)$$

$$= b_1 \begin{bmatrix} 5 \\ 1 \end{bmatrix} + b_2 \begin{bmatrix} 0 \\ 7 \end{bmatrix} + b_3 \begin{bmatrix} 5 \\ -6 \end{bmatrix},$$

so

$$L(\mathbf{x}) = \begin{bmatrix} 5b_1 + 5b_3 \\ b_1 + 7b_2 - 6b_3 \end{bmatrix}. \quad (1)$$

(c) To find $L\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right)$, we first have (verify)

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}; \quad \text{hence} \quad \left[\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right]_{S'} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Then, using $b_1 = 1$, $b_2 = 0$, and $b_3 = 2$ in Equation (1), we obtain

$$L \left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 15 \\ -11 \end{bmatrix}.$$



Similarity

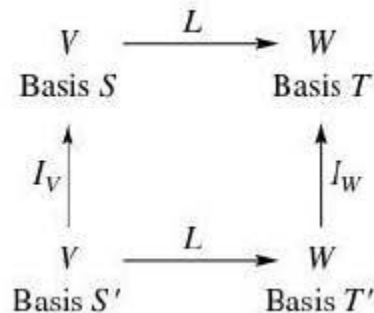
Theorem

Let $L: V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W . Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $S' = \{\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_n\}$ be ordered bases for V , with transition matrix P from S' to S ; let $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ and $T' = \{\mathbf{w}'_1, \mathbf{w}'_2, \dots, \mathbf{w}'_m\}$ be ordered bases for W with transition matrix Q from T' to T . If A is the representation of L with respect to S and T , then $Q^{-1}AP$ is the representation of L with respect to S' and T' .

Similarity

Theorem

Let $L: V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W . Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $S' = \{\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_n\}$ be ordered bases for V , with transition matrix P from S' to S ; let $T = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ and $T' = \{\mathbf{w}'_1, \mathbf{w}'_2, \dots, \mathbf{w}'_m\}$ be ordered bases for W with transition matrix Q from T' to T . If A is the representation of L with respect to S and T , then $Q^{-1}AP$ is the representation of L with respect to S' and T' .



Example 1. Let $V = \mathbb{R}^2$ and consider the bases

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$\mathbf{e}'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{e}'_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

is the transition matrix from the basis $B' = \{\mathbf{e}'_1, \mathbf{e}'_2\}$ to the basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$.

Let $L: R^3 \rightarrow R^2$ be defined by

$$L\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 + u_3 \\ u_2 - u_3 \end{bmatrix}.$$

Consider the ordered bases

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \text{and} \quad S' = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

for R^3 , and

$$T = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} \quad \text{and} \quad T' = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$$

for R^2 . We can establish (verify) that $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ is the representation of L with respect to S and T .

The transition matrix P from S' to S is the matrix whose j th column is the coordinate vector of the j th vector in the basis S' with respect to S . Thus

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \text{ and the transition matrix } Q \text{ from } T' \text{ to } T \text{ is } Q = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

$$\text{Now } Q^{-1} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}. \text{ (We could also obtain } Q^{-1} \text{ as the transition matrix}$$

from T to T' .) Then the representation of L with respect to S' and T' is

$$B = Q^{-1}AP = \begin{bmatrix} 1 & \frac{3}{2} & 2 \\ 0 & -\frac{1}{2} & -1 \end{bmatrix}.$$

On the other hand, we can compute the representation of L with respect to S' and T' directly. We have

$$L\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \text{so } \left[L\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) \right]_{T'} = \begin{bmatrix} 1 \\ 0 \end{bmatrix};$$

$$L\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{3}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \text{so } \left[L\left(\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}\right) \right]_{T'} = \begin{bmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{bmatrix};$$

$$L\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \text{so } \left[L\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) \right]_{T'} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Then the representation of L with respect to S' and T' is

$$\begin{bmatrix} 1 & \frac{3}{2} & 2 \\ 0 & -\frac{1}{2} & -1 \end{bmatrix},$$

which agrees with our earlier result. ■

Corollary

Let $L: V \rightarrow V$ be a linear operator on an n -dimensional vector space. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ and $S' = \{\mathbf{v}'_1, \mathbf{v}'_2, \dots, \mathbf{v}'_n\}$ be ordered bases for V with transition matrix P from S' to S . If A is the representation of L with respect to S , then $P^{-1}AP$ is the representation of L with respect to S' . ■

DEFINITION

If A and B are $n \times n$ matrices, we say that B is **similar** to A if there is a nonsingular matrix P such that $B = P^{-1}AP$.

Theorem

Let V be any n -dimensional vector space and let A and B be any $n \times n$ matrices. Then A and B are similar if and only if A and B represent the same linear transformation $L: V \rightarrow V$ with respect to two ordered bases for V . ■

Theorem

Let V be any n -dimensional vector space and let A and B be any $n \times n$ matrices. Then A and B are similar if and only if A and B represent the same linear transformation $L: V \rightarrow V$ with respect to two ordered bases for V . ■

Let $L: R_3 \rightarrow R_3$ be defined by

$$L \left(\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} \right) = \begin{bmatrix} 2u_1 - u_3 & u_1 + u_2 - u_3 & u_3 \end{bmatrix}.$$

Let $S = \{[1 \ 0 \ 0], [0 \ 1 \ 0], [0 \ 0 \ 1]\}$ be the natural basis for R_3 . The representation of L with respect to S is

$$A = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now consider the ordered basis

$$S' = \{[1 \ 0 \ 1], [0 \ 1 \ 0], [1 \ 1 \ 0]\}$$

for R_3 . The transition matrix P from S' to S is

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}; \quad \text{moreover, } P^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

Then the representation of L with respect to S' is

$$B = P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

The same result can be calculated directly (verify). The matrices A and B are similar. ■

Theorem

If A and B are similar $n \times n$ matrices, then $\text{rank } A = \text{rank } B$.

- Ex : (Finding a transformation matrix relative to nonstandard bases)

Let $T: R^2 \rightarrow R^2$ be a L.T. defined by

$$T(x_1, x_2) = (x_1 + x_2, 2x_1 - x_2)$$

Find the matrix of T relative to the basis

$$B = \{(1, 2), (-1, 1)\} \text{ and } B' = \{(1, 0), (0, 1)\}$$

Sol:

$$T(1, 2) = (3, 0) = 3(1, 0) + 0(0, 1)$$

$$T(-1, 1) = (0, -3) = 0(1, 0) - 3(0, 1)$$

$$[T(1, 2)]_{B'} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad [T(-1, 1)]_{B'} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

the transformation matrix T relative to B and B'

$$A = \begin{bmatrix} [T(1, 2)]_{B'} & [T(-1, 1)]_{B'} \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix}$$

Now use the matrix A to find $T(\mathbf{v})$, where $\mathbf{v} = (2, 1)$

$$\mathbf{v} = (2, 1) = 1(1, 2) - 1(-1, 1)$$

$$B = \{(1, 2), (-1, 1)\}$$

$$\Rightarrow [\mathbf{v}]_B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow [T(\mathbf{v})]_{B'} = A[\mathbf{v}]_B = \begin{bmatrix} 3 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\Rightarrow T(\mathbf{v}) = 3(1, 0) + 3(0, 1) = (3, 3)$$

$$B' = \{(1, 0), (0, 1)\}$$

- Check: $T(2, 1) = (2 + 1, 2(2) - 1) = (3, 3)$

• Notes:

(1) In the special case where $V = W$ and $B = B'$,
the matrix A is called the matrix of T relative to the basis B

(2) $T : V \rightarrow V$: the identity transformation

$B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{L}, \mathbf{v}_n\}$: a basis for V

$\Rightarrow$ the matrix of T relative to the basis B

$$A = \left[[\mathbf{T}(\mathbf{v}_1)]_B, [\mathbf{T}(\mathbf{v}_2)]_B, \mathbf{L}, [\mathbf{T}(\mathbf{v}_n)]_B \right] = \begin{bmatrix} 1 & 0 & \mathbf{L} & 0 \\ 0 & 1 & \mathbf{L} & 0 \\ \mathbf{M} & \mathbf{M} & \mathbf{O} & \mathbf{M} \\ 0 & 0 & \mathbf{L} & 1 \end{bmatrix} = I_n$$

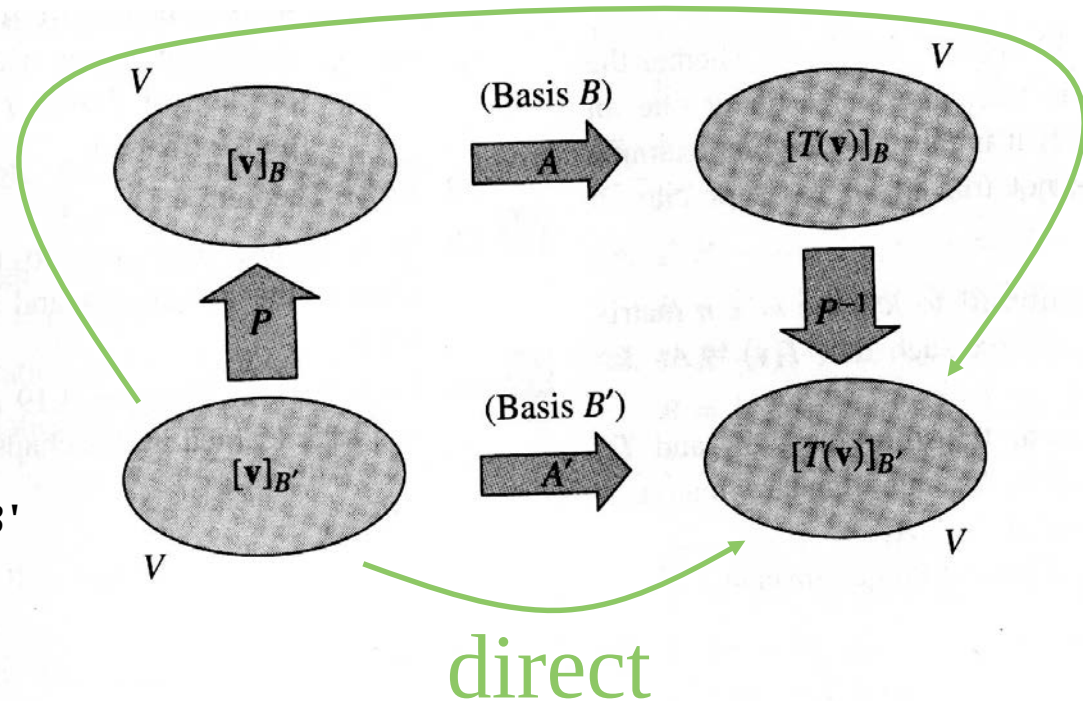
- Two ways to get $[v]_{B'}$ from $[T(v)]_{B'}$ to $[v]_{B'}$:
 (1) direct
 (2) indirect

$$A'[v]_{B'} = [T(v)]_{B'}$$

(2) indirect

$$P^{-1}AP[v]_{B'} = [T(v)]_{B'}$$

$$\Rightarrow \mathbf{A'}_{B'} = \mathbf{P}^{-1}_{B'B} \mathbf{A}_B \mathbf{P}_{BB'}$$



Example 1. Let $V = \mathbb{R}^2$ and consider the bases

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

and

$$\mathbf{e}'_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{e}'_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Then

$$P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

is the transition matrix from the basis $B' = \{\mathbf{e}'_1, \mathbf{e}'_2\}$ to the basis $B = \{\mathbf{e}_1, \mathbf{e}_2\}$.

Example 2. We continue Example 1. If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear operator given by $T(x, y) = (x + y, x - y)$, then its matrix in the basis B is

$$[T]_B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

and

$$[T]_{B'} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & -2 \end{bmatrix}$$

- **Ex** Find the transformation matrix A' for $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
with $T(x_1, x_2) = (2x_1 - 2x_2, -x_1 + 3x_2)$
relative to the basis $B' = \{(1, 0), (1, 1)\}$

Sol:

$$(I) A' = \left[[T(1, 0)]_{B'} \quad [T(1, 1)]_{B'} \right]$$

$$T(1, 0) = (2, -1) = \mathbf{3}(1, 0) - \mathbf{1}(1, 1) \Rightarrow [T(1, 0)]_{B'} = \begin{bmatrix} \mathbf{3} \\ \mathbf{-1} \end{bmatrix}$$

$$T(1, 1) = (0, 2) = \mathbf{-2}(1, 0) + \mathbf{2}(1, 1) \Rightarrow [T(1, 1)]_{B'} = \begin{bmatrix} \mathbf{-2} \\ \mathbf{2} \end{bmatrix}$$

$$\Rightarrow A' = \left[[T(1, 0)]_{B'} \quad [T(1, 1)]_{B'} \right] = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$$

(II) Standard matrix for T (i.e., the transformation matrix of T relative to $B = \{(1, 0), (0, 1)\}$)

$$A = [T(1, 0) \quad T(0, 1)] = \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} \quad \text{with} \quad T(x_1, x_2) = (2x_1 - 2x_2, -x_1 + 3x_2)$$

The transition matrix from B' to B : $P = \begin{bmatrix} [(1, 0)]_B & [(1, 1)]_B \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

The transition matrix from B to B' : $P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

The transformation matrix of T relative $B' \{(1, 0), (1, 1)\}$

$$A' = P^{-1}AP = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix}$$

- **Similar matrix:**

For square matrices A and A' of order n , A' is said to be similar to A if there exist **an invertible matrix P** such that

$$A' = P^{-1}AP$$

- **Thm 4.12:** (Properties of similar matrices)

Let A , B , and C be square matrices of order n .

Then the following properties are true.

- (1) A is similar to A .
- (2) If A is similar to B , then B is similar to A .
- (3) If A is similar to B and B is similar to C , then A is similar to C .

Pf:

$$(1) A = I_n A I_n$$

$$(2) A = P^{-1}BP \Rightarrow PAP^{-1} = P(P^{-1}BP)P^{-1}$$

$$PAP^{-1} = B \Rightarrow Q^{-1}AQ = B \quad (Q = P^{-1})$$

- Ex : (A comparison of two matrices for a linear transformation)

Suppose $A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ is the matrix for $T : \mathbf{R}^3 \rightarrow \mathbf{R}^3$ relative

to the standard basis B . Find the matrix for T relative to the basis

$$B' = \{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}$$

Sol:

The transition matrix P from B' to the standard basis B is

$$P = \begin{bmatrix} [(1, 1, 0)]_B & [(1, -1, 0)]_B & [(0, 0, 1)]_B \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The matrix of T relative to B' :

$$\begin{aligned} A' = P^{-1}AP &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}_{B \rightarrow B'} \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}_B \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{B' \rightarrow B} \\ &= \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \textit{diagonal matrix} \end{aligned}$$